Xuzhong Wang

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Education

Columbia University, Fu Foundation School of Engineering *May 2028 (expected)*
B.S. in Computer Science

College of William & Mary *May 2026*
B.S. in Mathematics and Computer Science, Honors in Mathematics *GPA: 3.99/4.0*

Publications & Preprints

†*Authors listed alphabetically (mathematics convention).* **Equal contribution.*

1. **X. Wang**, M. Jiang, T. Nair, G. Bhusal, Y. Zhang, H. Chen. "Towards Reliable, Generalizable, and Specific In-Context Knowledge Editing via Multi-Objective Reinforcement Learning." *Under review, ACL ARR 2026 (May).*
2. T. Nair, M. Jiang, **X. Wang**, A. Gao, G. Zhou, H. Chen, Y. Zhang. "Adaptive Context Selection for In-Context Learning via Steering Prediction." *Under review, ACL ARR 2026 (May).*
3. S. Al Jannat, Y. Li, M. He, **X. Wang**, H. Zhao, J. Sun, D. Xu, E. Xu, Y. Sun. "Akita: A High Usability Simulation Framework for Computer Architecture." *arXiv:2604.28073, 2026.*
4. †D. Cranston, **Xuzhong Wang**, G. Yu. "List-Recoloring of Graphs with Small Maximum Average Degree." *Submitted, 2026.*

Research Experience

Undergraduate Researcher, D3i Lab May 2025 – Present
Advisors: Haipeng Chen, Yanfu Zhang

- Designed and implemented a **multi-objective RL framework** to train an auxiliary retriever policy for in-context knowledge editing.
- Mitigated reward hacking observed in prior work via constrained reward formulation, improving editing specificity by ∼**10%** over existing baselines.
- Currently investigating **robust LLM unlearning** and developing improved **multi-objective RL algorithms** for LLM post-training.

Undergraduate Honors Thesis, Graph Theory March 2025 – May 2026
Advisors: Daniel W. Cranston, Gexin Yu

- Introduced the **Generalized Extension Lemma**, strengthening the Extension Lemma for bounding recoloring sequences from a single vertex to arbitrary subgraphs.
- Developed a **global discharging argument** that exploits global graph structure, extending beyond classical local discharging to obtain sharper density bounds.
- **Proved** that 5-list recoloring has linear diameter for $\mathrm{mad}(G) < 3 - \varepsilon$; **improve the linear bound** for 6-list recoloring from 3.4 to $3.75 - \varepsilon$.

Undergraduate Researcher, Scalable Architecture Lab May 2024 – November 2025
Advisor: Yifan Sun

- Implemented **data recording tool** in Go and SQLite for simulation data tracking, enabling efficient read/write support for experiments.
- Maintain Akita, a computer architecture simulator engine, enhancing developer experience through **event-driven design** and **smart ticking** for simulation.

Professional Experience

Machine Learning Engineer Intern, AnytimeAI June 2026 – Present
White Plains, NY

- Design **LLM agent pipelines** for cross-document correlation analysis of legal documents.
- Built an agent routing/advisor tool that reduces model inference costs by ∼**60%**.
- Evaluated a multimodal ensemble for extracting and analyzing tabular data from plain text, improving task performance by ∼**8%**.

Machine Learning Engineer Intern, Bestlink Technology May 2024 – July 2024
Nanjing, Jiangsu

- Designed an **data retrieval pipeline** with OpenCV to extract QA pairs from images.
- Partnered with senior engineers to architect a **LangChain-based LLM agent workflow** powering a question-answering tool for consulting clients.

Talks & Presentations

- Poster: "List-Recoloring of Graphs with Small Maximum Average Degree." Joint Mathematics Meetings (JMM), Washington D.C., January 2026.

Grants & Awards

- Charles Center Summer Research Grant, College of William & Mary, 2025, 2026.
- Semester Research Grant, College of William & Mary, 2025.
- Cissy Patterson Grant for Research, College of William & Mary, 2025.
- JMM Travel Grant, Department of Mathematics, College of William & Mary, 2026.